

traditional techniques in tasks such as image classification, object detection, and transfer learning.

In conclusion, rapid advancements in AI and deep learning models have propelled visual data analysis to new heights. The benefits of deep learning, such as improved accuracy and flexibility, along with the emergence of innovative models like vision transformers, CLIP, SimCLR, and MoCo, have transformed traditional techniques. These advancements have opened doors to groundbreaking applications across various domains. However, the following challenges still require further research.

Model interpretability:

Interpretability is the degree to which a human can understand the cause of a decision. The higher the interpretability of a machine learning model, the easier it is for someone to comprehend why certain decisions or predictions have been made. Model interpretability of deep neural networks (DNNs) has always been a limiting factor for use cases requiring explanations of the features involved in modelling. This is the case for many industries such as financial services.

Robustness and adversarial defence: Adversarial attacks have revealed vulnerabilities in deep learning models. The community has identified the need for robust models that can withstand adversarial perturbations and maintain performance under different conditions. Efforts have been made to develop defences against adversarial attacks, study model robustness, and improve generalisation capabilities.

Data efficiency and generalisation: Deep learning models often require large amounts of labelled data for effective training. The need for data efficiency and better generalisation has been

Table 1: The latest DL models and what they replace

Traditional techniques	Replaced by
CNNs, feature extraction methods, region-based CNNs, image classification models, object detection models, GANs	Vision transformers
Image captioning models, visual question answering models, image retrieval models	CLIP
Transfer learning techniques	Vision transformers, CLIP, SimCLR
Self-supervised learning methods	SimCLR, MoCo, vision transformers
Image segmentation models, video understanding models	Vision transformers, CLIP
Data augmentation methods	SimCLR, MoCo
Few-shot learning methods, anomaly detection models, video action recognition models, image compression techniques	Vision transformers, CLIP

recognised to overcome the limitations of data availability and reduce reliance on labelled examples.

Multimodal and cross-modal learning: Multimodal data is data that uses multiple modalities. Combining and integrating information from multiple modalities, such as text, images, video, and audio, has gained

importance for tasks like multimedia understanding, sentiment analysis, and human-computer interaction.

Deep learning is reshaping the way we see and understand the world around us. The future holds limitless possibilities as we continue to unlock the potential of AI in unravelling the mysteries of visual data. END 🐧

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